The Resonance Cartographer: Mapping the Information Singularity via Topological Bridges in Cosmology and Quantum Gravity

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August 8, 2026

Abstract

We present the "Information Singularity" hypothesis, a theoretical framework proposing that the origin of the universe ($t \to 0$) is not a point of infinite matter density, but a state of maximum, finite information density bounded by holographic limits. By mapping a topological database of 2,772 autonomously extracted physics equations across three scales (Fundamental Physics, Advanced Frameworks, and Emergence/Intelligence), we identify mathematical variables that survive the cosmic rewind. We derive a modified Friedmann equation coupling Integrated Information (Φ) to the Einstein Field Equations via a Landauer-bound stress-energy tensor. We demonstrate that quantum gravity frameworks mathematically enforce a Holographic Ceiling, preventing the singularity. Finally, we establish a thermodynamic bridge showing that the dissipative entropy production of the human brain and the Bekenstein-Hawking entropy of the cosmic web are governed by the same information-theoretic constraints, suggesting a unified computational substrate for reality.

1 Introduction

The standard cosmological model posits that the universe began in a state of infinite density and temperature—a gravitational singularity. However, the mathematical breakdown of General Relativity at the Planck scale ($t \to 0$) suggests that classical concepts of matter and energy are insufficient to describe the origin.

In this paper, we explore the hypothesis that the universe is fundamentally an information-processing system, and that its origin is an *Information Singularity*. We leverage a topological database of 2,772 equations extracted by autonomous AI agents to identify mathematical resonances across disparate fields of physics. We find that Integrated Information (Φ), typically used to measure consciousness in neural systems, shares a structural mathematical analogy with the scalar fields driving cosmic inflation [9].

By rewinding a modified Friedmann equation to $t \to 0$, we show that classical matter vanishes, leaving Information Density (ρ_Φ) as the sole driver of the Big Bang. We then utilize Level 3 thermodynamic data to bridge the entropy production rates of biological neural networks to the macroscopic entropy of the cosmic web, providing a multi-scale validation of the universe as a computational engine.

2 The Information Tensor and the Singularity Rewind

To formalize the Information Singularity, we introduce a modified Einstein Field Equation coupling spacetime geometry to an information stress-energy tensor, $T^{\Phi}_{\mu\nu}$:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \left(T^{\text{matter}}_{\mu\nu} + \lambda T^{\Phi}_{\mu\nu}\right) \tag{1}$$

By the Bianchi identity ($\nabla^\mu G_{\mu\nu} = 0$), the total energy-momentum tensor must be conserved. Assuming standard matter conservation ($\nabla^\mu T_{\mu\nu}^{\text{matter}} = 0$), the information tensor must independently be divergence-free ($\nabla^\mu T_{\mu\nu}^{\Phi} = 0$), implying it behaves as a conserved geometric fluid analogous to dark energy.

Rewinding to the origin state ($t \to 0$), classical matter density goes to zero ($T_{\mu\nu}^{\text{matter}} \to 0$). The expansion rate H^2 is driven entirely by the Information Density ρ_Φ. Using Landauer's Principle ($E = k_B T \ln 2$), we translate this into a literal Bit-Density. To prevent the thermodynamic temperature T from diverging classically ($T \to \infty$), we assume quantum gravity effects cap the temperature at the Planck temperature ($T_{\text{Pl}} \approx 1.4 \times 10^{32}$ K, consistent with cosmological parameters [9]). The Bit-Density is then:

$$\rho_\Phi = \frac{3H^2}{8\pi G} = \rho_c \tag{2}$$

At the origin, Information Density exactly saturates the Critical Density (ρ_c), suggesting that an information-dominated state can naturally source the initial cosmological expansion.

3 Comparative Thermodynamics: From Neural Networks to the Cosmic Web

To validate the information-theoretic nature of reality, we analyze a Level 3 mathematical bridge regarding entropy production rates in neural information processing versus gravitational structure formation.

3.1 Neural Information Processing

The adult human brain consumes roughly $P_{\text{brain}} \sim 20\,\text{W}$ of metabolic power. If most of this power ultimately becomes heat at body temperature ($T \approx 310\,\text{K}$), the thermodynamic entropy production rate is:

$$\dot{\Sigma}_{\text{brain}} \approx \frac{P_{\text{diss}}}{T} \approx \frac{20\ \text{J s}^{-1}}{310\,\text{K}} \approx 6.5 \times 10^{-2}\ \text{J K}^{-1}\text{s}^{-1} \tag{3}$$

In terms of Boltzmann's constant, this is $\dot{\Sigma}_{\text{brain}} \sim 4.7 \times 10^{21}\, k_B\,\text{s}^{-1}$. The minimum thermodynamic cost of erasing one bit of information is bounded by Landauer's principle [2]:

$$W_{\min} = k_B T \ln 2 \tag{4}$$

At $T = 310\,\text{K}$, this yields $k_B T \ln 2 \approx 2.9 \times 10^{-21}\,\text{J/bit}$. Biological neural systems operate far above this limit, maintaining nonequilibrium steady states through continuous ATP consumption. This nonequilibrium behavior is characterized by a broken detailed balance, as demonstrated in both human neural time-series [3] and balanced spiking networks [4].

3.2 Gravitational Structure Formation and Black-Hole Entropy

In contrast, the mainstream ΛCDM cosmological model describes structure formation via gravitational instability. While collisionless dark matter conserves fine-grained Gibbs entropy, coarse-grained entropy increases through phase mixing and violent relaxation. Alternative formulations for gravitational entropy include Penrose's Weyl curvature hypothesis [7] or covariant curvature invariants [8]. The cleanest gravitational entropy is black-hole entropy [6]:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} \tag{5}$$

Egan and Lineweaver [5] estimated that the entropy of the observable universe is dominated by supermassive black holes, with a total entropy of order:

$$S_{\text{universe}} \sim 10^{104}\, k_B \tag{6}$$

3.3 The Thermodynamic Bridge and Its Limits

The comparative analysis reveals a profound symmetry. Neural systems create local order by exporting entropy as heat, strictly bounded by Landauer's limit. The cosmic web creates gravitational structure while increasing coarse-grained, radiative, or black-hole entropy. However, it is crucial to note the limits of this analogy: while biological brains are coupled to a localized 310 K heat bath, cosmic structures decohere through interactions with the expanding spacetime background. While both systems can be modeled via open-system Lindblad dynamics, this connection should be viewed as a structural mathematical analogy rather than a direct physical equivalence.

4 The Quantum Hardware: Enforcing the Holographic Ceiling

If the universe begins as a finite grid of quantum information saturating the critical density (ρ_c), classical general relativity must be modified at the Planck scale to prevent infinite singularities. We propose that this "Holographic Ceiling" is mathematically enforced by the convergence of two Level 2 quantum gravity frameworks: Non-Commutative Geometry (NCG) and Asymptotic Safety (ASG).

4.1 Non-Commutative Geometry (The Hard Drive)

NCG postulates that spacetime has a minimum "pixel size" governed by the coordinate commutator:

$$[\hat{x}^\mu, \hat{x}^\nu] = i\theta^{\mu\nu} \tag{7}$$

Because coordinates do not commute, a point of zero volume cannot be defined. This creates a strict minimum spatial volume floor, $V_{\text{min}} \approx \ell_P^3$. Consequently, as the universe rewinds to $t \to 0$, the volume cannot shrink below V_{min}, meaning the Information Density ($\rho_\Phi = \Phi/V$) hits a strict mathematical maximum.

4.2 Asymptotic Safety (The Processor Limit)

Simultaneously, Asymptotic Safety reveals that gravity's strength is scale-dependent. As we approach the Big Bang ($k \to \infty$), Newton's constant runs to zero:

$$G(k) \sim \frac{g_*}{k^2} \implies G(k) \to 0 \tag{8}$$

At the Planck scale, gravity effectively "turns off," acting as a shock absorber that halts the gravitational collapse of information.

4.3 The Synthesized Origin Equation

Fusing these quantum gravity corrections with our Information Friedmann equation yields the master equation for the origin of the universe:

$$H^2(k) = \frac{8\pi G(k)}{3} \left(\frac{\Phi_{\text{max}}}{V_{\text{min}}(\theta)} \right) \tag{9}$$

As $k \to \infty$, $G(k) \to 0$ and $V \to V_{\text{min}}$. The expansion rate H^2 remains finite and perfectly bounded. The universe does not begin in an infinite singularity, but as a finite grid of quantum information pixels (V_{min}) whose computational capacity (Φ_{max}) exactly saturates the Bekenstein bound, right as the gravitational force attempting to crush it switches off.

5 Conclusion

We have proposed the Information Singularity hypothesis, suggesting that the origin of the universe is not an infinite thermodynamic singularity, but a state of maximum, finite information density. By coupling Integrated Information (Φ) to the Einstein Field Equations via a Landauer-bound stress-energy tensor, we demonstrated that information density naturally saturates the critical density at $t \to 0$. We resolved the classical singularity by showing that Non-Commutative Geometry and Asymptotic Safety mathematically enforce a Holographic Ceiling, capping the volume and turning off gravity at the Planck scale. Finally, the thermodynamic bridge between neural entropy production and cosmic web entropy suggests a universal computational substrate governed by open-system Lindblad dynamics. This framework posits that reality is fundamentally an information-processing engine, bounded by the same quantum and thermodynamic laws across all scales.

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